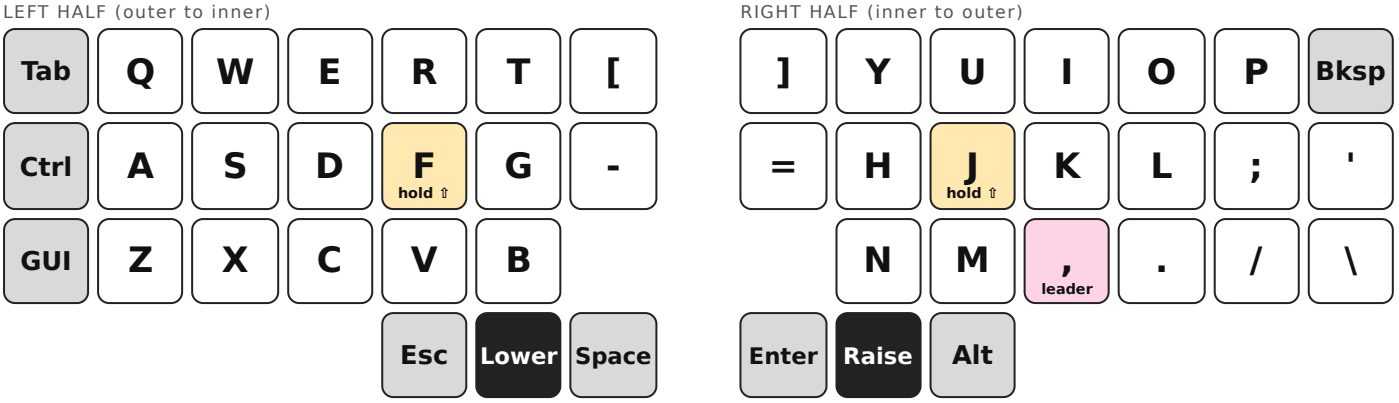
HW46 key map

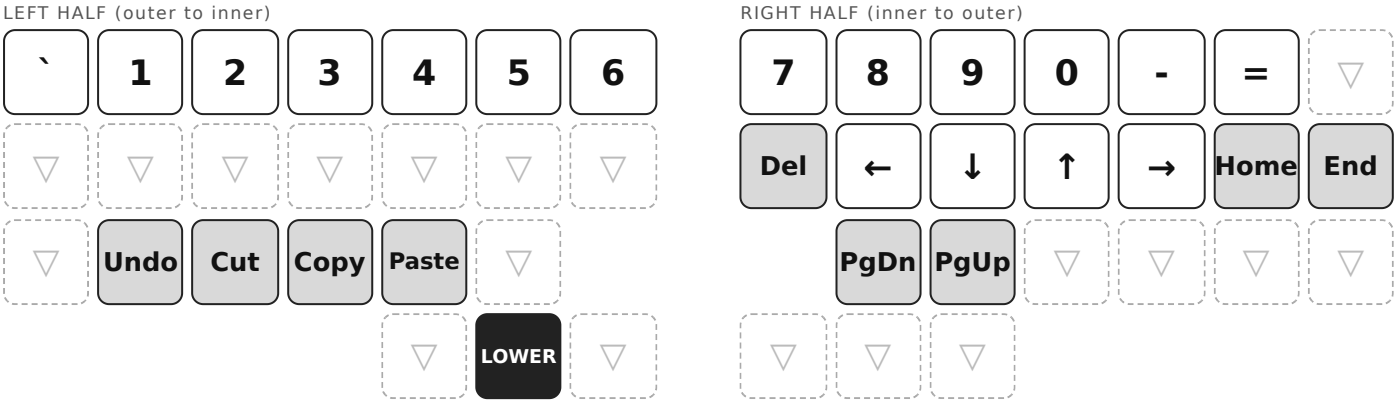
46-key wireless split, top view. Left half = Bluetooth name HW46, right half = HW46-R. Same layers on both halves.

BASE (QWERTY)



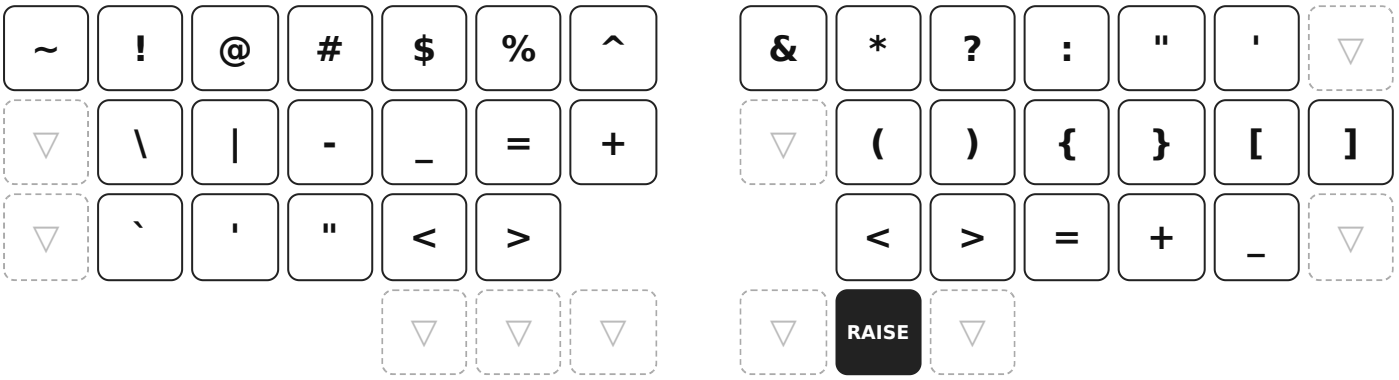
LOWER: vim arrows, numbers, clipboard

hold the LEFT thumb middle key



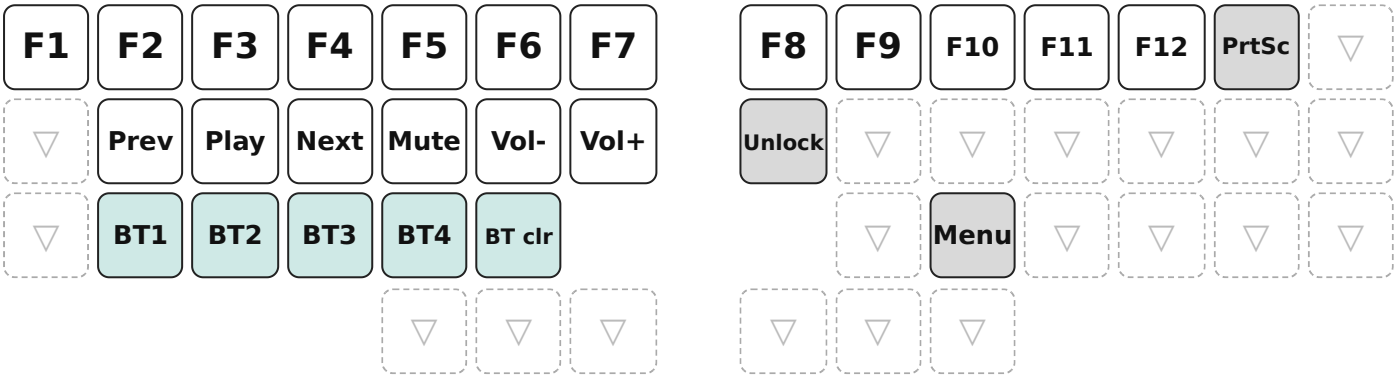
RAISE: programming symbols

hold the RIGHT thumb middle key



ADJUST: Lower + Raise held together

F-keys, media, Bluetooth, Studio unlock



F, J: tap = letter, hold = Shift (other hand's keys) · Leader (,) · Layer key · Modifier / special · Bluetooth · Passes through to the layer below

HW46 wiring and quick reference

Same wiring on both halves. Pins as printed on the board (GPIO names).

Stacks (columns of keys)

Stack	Pin	
S1	**022**	inner (next to other half)
S2	**024**	
S3	**100**	
S4	**011**	
S5	**104**	
S6	**106**	
S7	**115**	outer edge

Cross wires (rows of keys)

Pin	Serves	
031	Row 1 (top)	7 keys
002	Row 2 (home row)	7 keys
111	Row 3 (bottom)	6 keys, no inner key
029	Thumb row	3 keys, inner three columns

Free for the trackpad: **017 020 113 010**

Which key sits at each crossing

LEFT HALF

	031 Row 1	**002** Row 2	**111** Row 3	**029** Thumb
S1 022	[	-		Space
S2 024	T	G	B	Lower
S3 100	R	F	V	Esc
S4 011	E	D	C	
S5 104	W	S	X	
S6 106	Q	A	Z	
S7 115	Tab	Ctrl	GUI	

RIGHT HALF

	031 Row 1	**002** Row 2	**111** Row 3	**029** Thumb
S1 022	]	=		Enter
S2 024	Y	H	N	Raise
S3 100	U	J	M	Alt
S4 011	I	K	,	
S5 104	O	L	.	
S6 106	P	;	/	
S7 115	Bksp	'	\	

Flash a half

1. Data cable, straight into the PC.
2. Double-tap reset quickly. Blue LED pulses, drive **NICENANO** appears.
3. Copy the **.uf2** onto it. The drive disappears when done. "Try again" is normal.
4. Keymap change: flash only **hw46_left**.
5. Pairing trouble: **settings_reset** on both, then **hw46_left** (left) and **hw46_right** (right).

Pair with the PC

1. Remove HW46 from the PC's Bluetooth list.
2. Tap reset on both halves.
3. Add **HW46** (not HW46-R).
4. Wait about 10 s; the right half joins the left.
5. A USB cable to a PC types over USB; a charger uses Bluetooth.

Handy facts

- Esc = left outer thumb. Ctrl = left home-row pinky. Hold F or J for Shift; **hold F + ;** gives **:**
- Paste = Lower + V (Shift+Insert). Copy = Lower + C (Ctrl+Insert). Cut = Lower + X (Shift+Delete). Undo = Lower + Z.
- Bluetooth profiles 1 to 4 and clear: Adjust layer (hold Lower + Raise), left half bottom row.
- ZMK Studio unlock key: Adjust layer, first key of the right half's home row.
- Battery lasts weeks; about 4 hours to charge from 27 %. Red LED on = charging.